

Ecr

1. Setup a highly available ecs cluster with load balancer and dynamic port mapping.

NOTE:

Your cluster should maintain atleast 6 tasks at any point of time and should be highly available across multiple AZ's.

Use the below image from deployment:
sabair0509/hiring-app:works

Goal in simple words:

Run your Docker app on many servers so that

- always **6 containers (tasks) are running**
- if one server dies → app still works
- traffic automatically spreads using **Load Balancer**
- ports auto-assigned (**dynamic port mapping**)
- multi-AZ = high availability

We will use:

- **ECS (EC2 launch type)** → runs containers
- **Auto Scaling Group** → keeps EC2 machines alive
- **Application Load Balancer (ALB)** → traffic distribution
- **Target group (dynamic ports)** → routing to containers
- **Service (desired count = 6)** → keeps 6 tasks always running

Image to deploy:

sabair0509/hiring-app:works

Big Architecture (understand first)

Internet → ALB → Target Group → ECS Service → Tasks → EC2 Instances → Across AZs

If

- one container stops → ECS starts new
- one EC2 dies → ASG launches new
- one AZ fails → other AZ still serving

That = **Highly Available**

Internet

↓

ALB (Port 80)

↓

Target Group (dynamic ports like 32768, 32769...)

↓

ECS Service (6 tasks)

↓

Containers (sabair0509/hiring-app:works → 8080)

↓

EC2 Instances (multiple AZs)

STEP 1 — Create ECS Cluster (EC2 capacity)

Go to:

ECS → Clusters → Create cluster

Cluster config

Setting	Value
Cluster name	hiring-cluster
Provisioning model	On-Demand
Instance type	t3.small (free tier ok)
Number of instances	3
Key pair	create or select
VPC	Default
Subnets	Select ALL subnets (important for HA)
Auto assign public IP	ENABLE

Create cluster ⌚ wait 3–5 min

▼ Infrastructure - *advanced* [Info](#)

Configure the manner of obtaining compute resources that will be used to host your application.

Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

☐ Fargate only
Serverless - you don't think about creating or managing servers. Great for most common workloads.

☐ Fargate and Managed Instances
Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.

☒ Fargate and Self-managed instances
Self-managed instances - you must ensure the instances are properly configured and scaled properly, and you have full control over the instances.

Auto Scaling group (ASG) | [Info](#)

☒ Create a new Auto Scaling group - *advanced*

☐ Use an existing Auto Scaling group

Provisioning model

Select a provisioning model for your instances

☒ On-demand
With on-demand instances, you pay for compute capacity by the hour, with no long-term commitments or upfront payments.

☐ Spot
Amazon EC2 Spot instances let you take advantage of unused EC2 capacity in the AWS cloud. [Learn more](#)

Container instance Amazon Machine Image (AMI)

Choose the Amazon ECS-optimized AMI for your instance.

Amazon Linux 2023

EC2 instance type

Choose based on the workloads you plan to run on this cluster.

t3.micro
x86_64 2 vCPU 1 GiB Memory

EC2 instance role

An instance role is used by Amazon EC2 instances to make AWS API requests. If you don't already have an instance IAM role created, we can create one for you.

Create default role

 [Create new instance profile](#)

Desired capacity

Specify the number of instances to launch in your cluster.

Then check:

ECS → Clusters → Infrastructure → **3 instances must show ACTIVE**

Dashboard	Find Instance by attribute or tag (case-sensitive)						All states
AWS Global View	running	X	Clear filters				
Events	☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
▼ Instances	☐	ECS Instance - ...	i-036e6eb82be58c8b5	Running	t3.micro	3/3 checks passed	View alarms +
Instances	☐	ECS Instance - ...	i-04d745556d07c575f	Running	t3.micro	3/3 checks passed	View alarms +
Instance Types	☐	ECS Instance - ...	i-0aff66bfd2921a9af	Running	t3.micro	3/3 checks passed	View alarms +

Clusters (1) Info							Last updated February 12, 2026, 20:13 (UTC+5:30)	Create cluster
Search clusters								
By default, we only load up to 1,000 clusters at a time.								
Cluster	Services	Tasks	Container instances	CloudWatch monitoring	Capacity provider strategy			
hiring-cluster	1	2 Pe... 5 Ru...	5 EC2	Default	ASG			

STEP 2 — Create Load Balancer

EC2 → Load Balancers → Create → Application Load Balancer

Basic

Field	Value
Name	my-alb
Scheme	Internet facing
IP	IPv4

Network

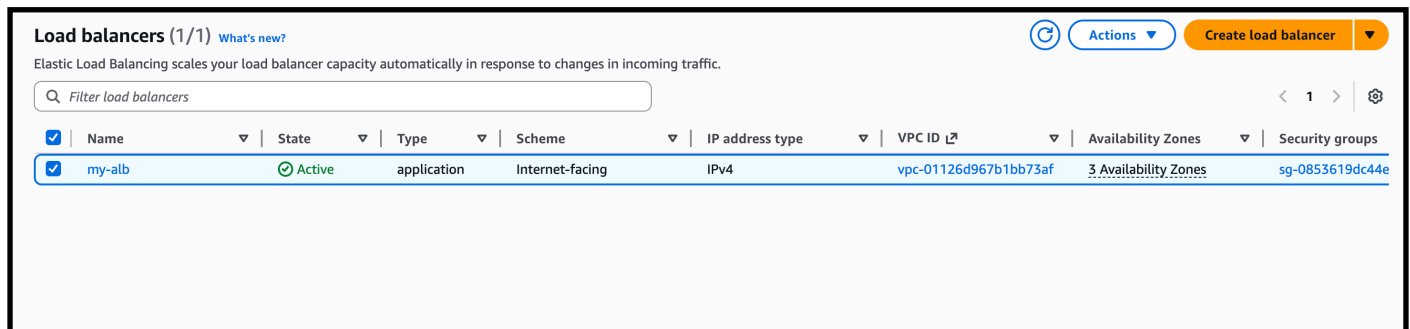
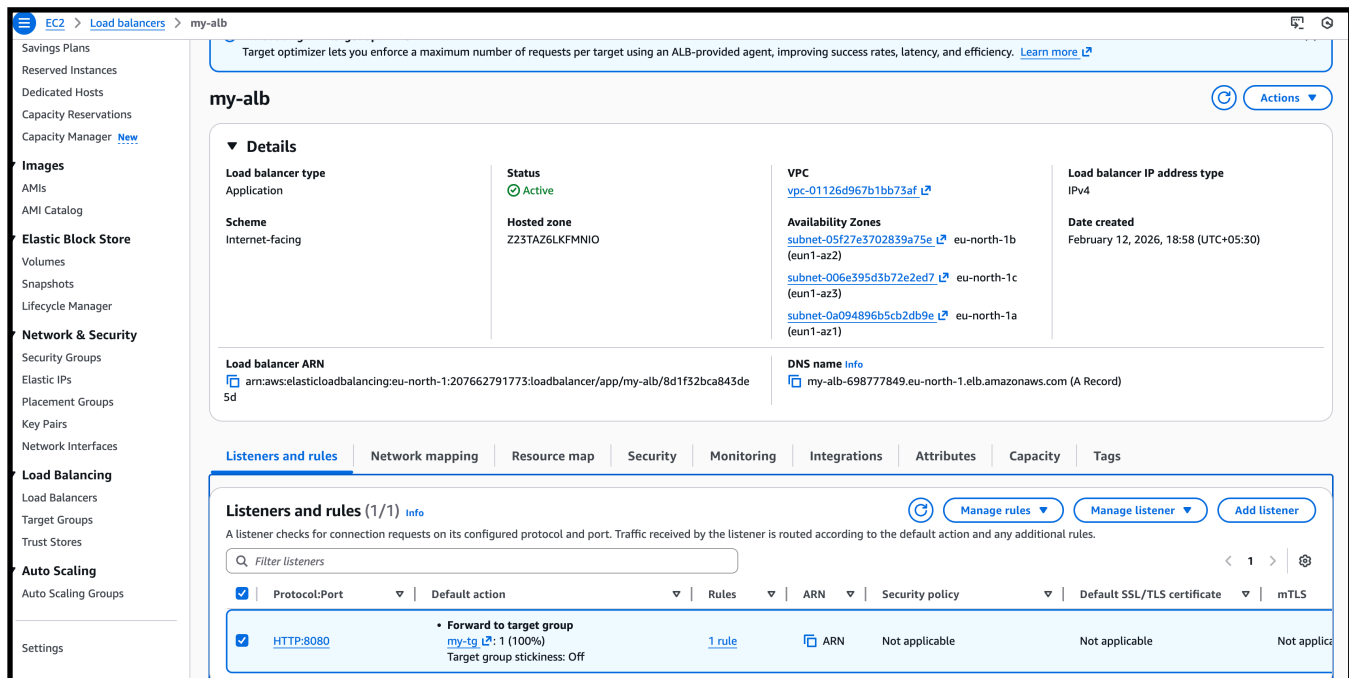
Select **ALL subnets** (very important for multi-AZ HA)

Security group

Create new:

Allow HTTP 80 from Anywhere

Allow Custom TCP 8080 from Anywhere (optional but safe)



STEP 3 — Create Target Group

EC2 → Target Groups → Create

Field	Value
Type	Instances
Protocol	HTTP
Port	8080
Name	my-tg
VPC	default

Health check

Setting	Value
Path	/hiring/
Success codes	200-499
Port	Traffic port

EC2 > Target groups									
Target groups (1/1) Info What's new?									
Filter target groups									
☒	Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID		
☒	my-tg	arn:aws:elasticloadbalancing...	8080	HTTP	Instance	my-alb	vpc-01126d967b1bb73af		

STEP 4 — Attach TG to ALB

Go to:
Load Balancer → Listeners → Add listener

Create:

Protocol	Port	Forward
HTTP	80	my-tg

EC2 > Load balancers > my-alb

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager New

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Settings

Target optimizer lets you enforce a maximum number of requests per target using an ALB-provided agent, improving success rates, latency, and efficiency. [Learn more](#)

my-alb

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

✔ Active

Hosted zone

Z23TAZ6LKFMNIO

VPC

vpc-01126d967b1bb73af

Availability Zones

subnet-05f27e3702839a75e eu-north-1b (eu1-az2)

subnet-006e395d3b72e2ed7 eu-north-1c (eu1-az3)

subnet-0a094896b5cb2db9e eu-north-1a (eu1-az1)

Load balancer IP address type

IPv4

Date created

February 12, 2026, 18:58 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:eu-north-1:207662791773:loadbalancer/app/my-alb/8d1f32bca843de5d

DNS name info

my-alb-698777849.eu-north-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1/1) info

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

Filter listeners

< 1 > ⚙

✔

Protocol:Port

▼

Default action

▼

Rules

▼

ARN

▼

Security policy

▼

Default SSL/TLS certificate

▼

mTLS

✔

[HTTP:8080](#)

• Forward to target group

my-tg 1 (100%)

Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

Not applicable

STEP 5 — Create Task Definition

ECS → Task Definitions → Create

Launch type

EC2
Network mode = bridge (IMPORTANT for dynamic port mapping)

Task size

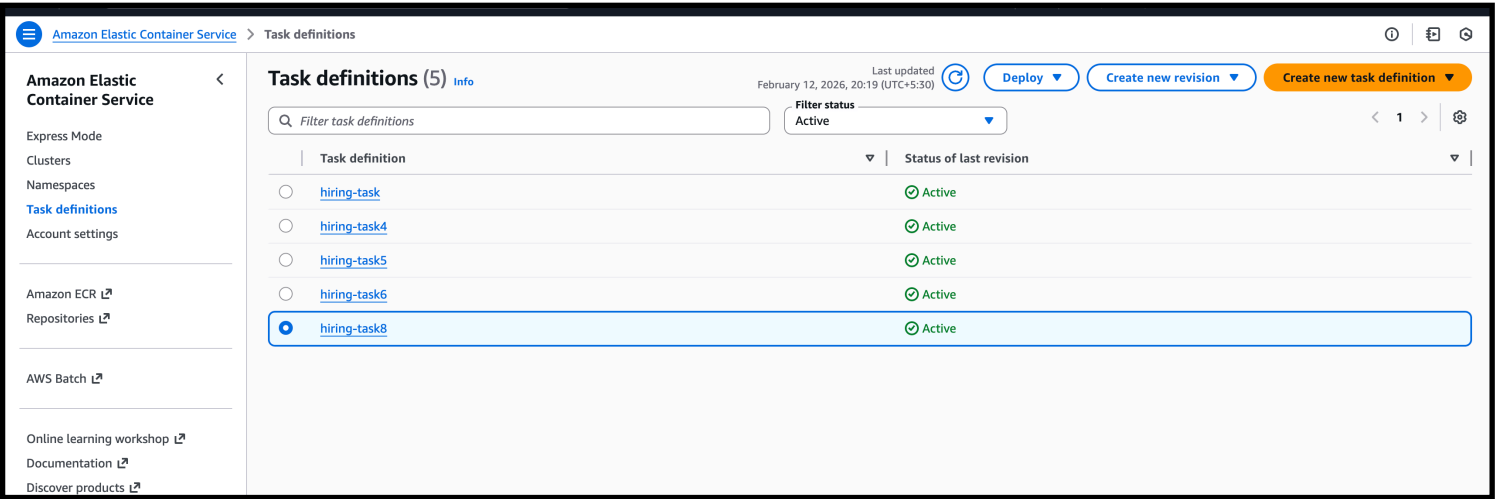
CP U	Memor y
256	512

Container

Field	Value
Name	app
Image	sabair0509/hiring-app:works
Container port	8080
Host port	0 ← ★ DYNAMIC PORT MAPPING

DO NOT add environment variables
DO NOT add volumes

Create

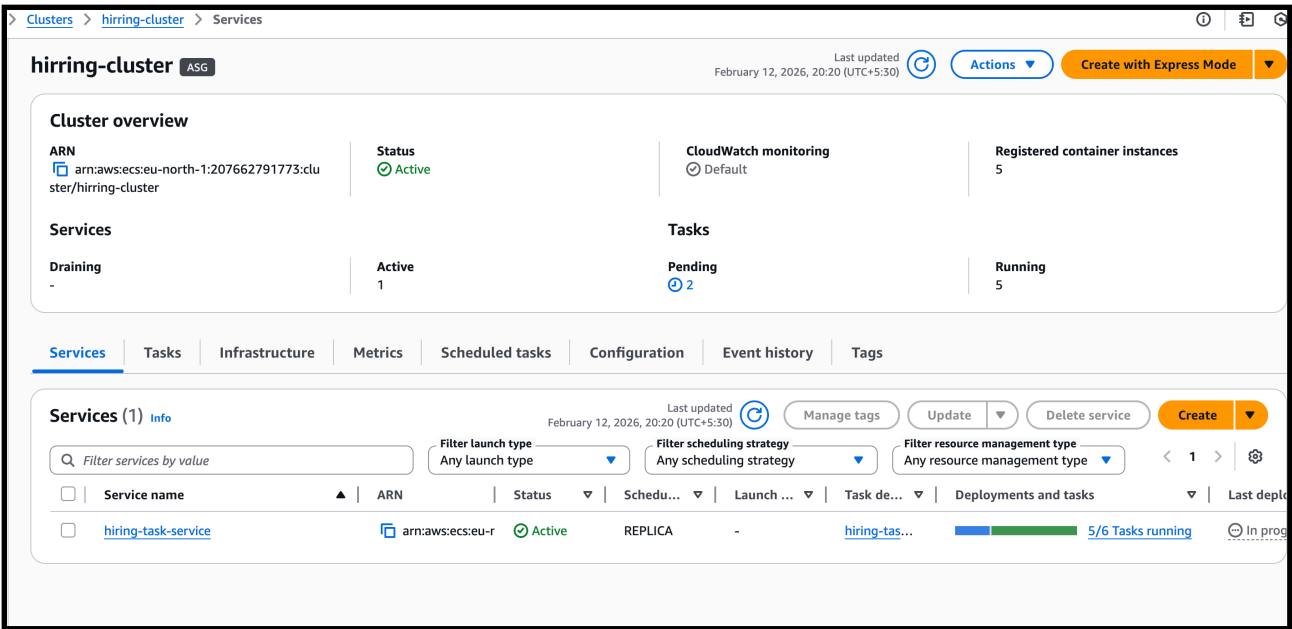


—> I have went through many modification this was the final modification which was success and running

STEP 6 — Create Service

ECS → Cluster → Services → Create

Setting	Value
Launch type	EC2
Task definition	hiring-task
Service name	hiring-service
Desired tasks	6



The screenshot shows the Amazon Elastic Container Service (ECS) console. The breadcrumb navigation at the top indicates the path: Amazon Elastic Container Service > Clusters > hiring-cluster > Services > hiring-task-service > Tasks. The left sidebar contains navigation links for Express Mode, Clusters, Namespaces, Task definitions, Account settings, Amazon ECR, Repositories, AWS Batch, Online learning workshop, Documentation, Discover products, and Subscriptions. The main content area displays a table of tasks for the 'hiring-task-service'. The first task, '2dffa92c2fad4ceba990fb...', is selected and highlighted. Below the task list, a section titled 'Containers for task 2dffa92c2fad4ceba990fbf49a3f79e' shows a table with one container named 'app' in a 'Running' state. The container's health status is 'Unknown'.

Task	Last status	Desired status	Task definition	Health status	Created at	Started by	Size
2dffa92c2fad4ceba990fb...	Running	Running	hiring-task6:1	Unknown	34 minutes ago	ecs-svc/13115784301...	34
3a0de915b7924001be70...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	33
61346f593311476b90ab...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	33
c580c9543e204b6091b5...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	31
ce8dbad08b6247e7950a...	Running	Running	hiring-task6:1	Unknown	33 minutes ago	ecs-svc/13115784301...	31
a39e347ed8a34d5a8fe58...	Provisioning	Running	hiring-task2	Unknown	20 minutes ago	ecs-svc/14539380324...	-

Container name	Container runtime ID	Image URI	Image Digest	Status	Health status	CPU	Memory hard/soft limit
app	ce3fd963d65442...	sabair050...	sha256:5...	Running	Unknown	0	- / -

—> container is running

Placement

Spread across AZ

Spread across Instance

Create service 🕒

STEP 6 — Verify

—> check your any server running and

—> docker images

—> docker ps



RESULT (Assignment completed)

You now have:

- ✓ Highly available (multi-AZ)
- ✓ 6 running tasks always
- ✓ Auto placement across instances
- ✓ Dynamic port mapping
- ✓ Load balancer routing traffic

